VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
INTERNATIONAL UNIVERSITY
SCHOOL OF ELECTRICAL ENGINEERING

# CAPSTONE DESIGN 2

## DESIGN AND FABRICATE SMART TEMPERATURE CONTROLLED MINI-FRIDGE

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HO CHI MINH CITY, VIETNAM
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Under the guidance and approval of the committee, and approved by its members, this report has been accepted as part of the requirements for Capstone Design 2.

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# HONESTY DECLARATION

All our's Group members - Group 6, we would like to declare that, apart from the acknowledged references, this Capstone Design 2 either does not use language, ideas, or other original material from anyone; or has not been previously submitted to any other educational and research programs or institutions. We fully understand that any content in this report that contradicts the above statement will automatically lead to the rejection of our Capstone Design 2 project by the School of Electrical Engineering, International University – Vietnam National University Ho Chi Minh City.

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# TURNITIN DECLARATION

Names of the students: Do Buu Phuoc, Nguyen Hong Hai Nguyen, Van Thao Minh, Nguyen The Hung, Nguyen Minh Quan.

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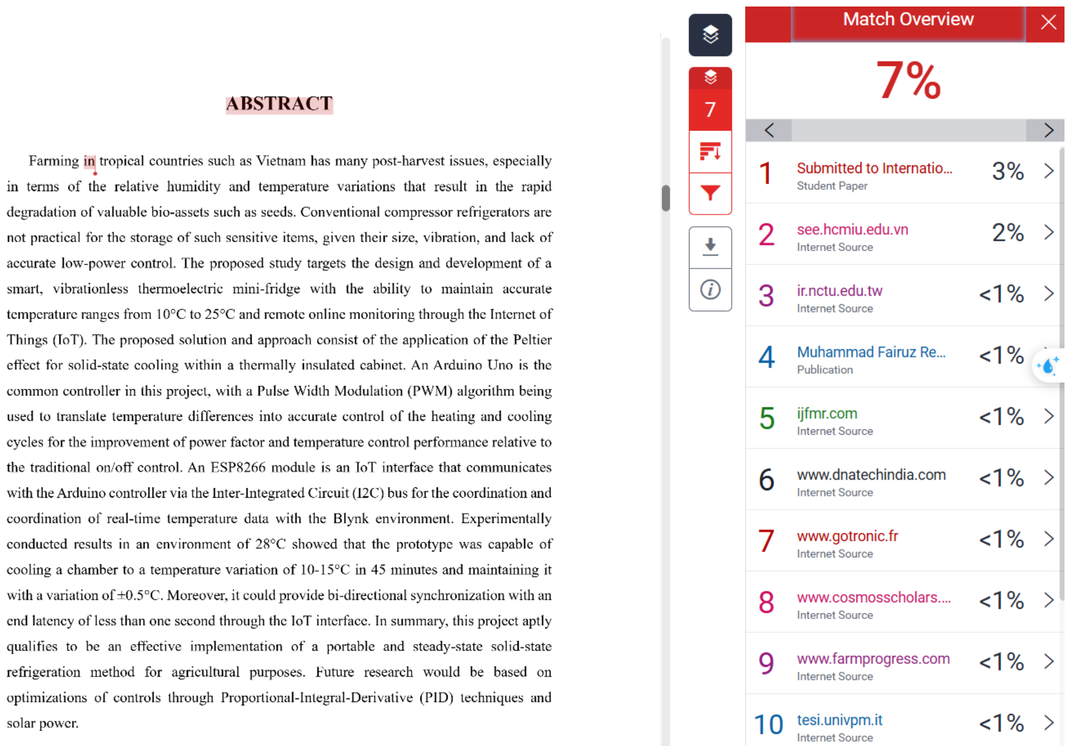

ABSTRACT

Farming in tropical countries such as Vietnam has many post-harvest issues, especially in terms of the relative humidity and temperature variations that result in the rapid degradation of valuable bio-assets such as seeds. Conventional compressor refrigerators are not practical for the storage of such sensitive items, given their size, vibration, and lack of accurate low-power control. The proposed study targets the design and development of a smart, vibrationless thermoelectric mini-fridge with the ability to maintain accurate temperature ranges from 10°C to 25°C and remote online monitoring through the Internet of Things (IoT). The proposed solution and approach consist of the application of the Peltier effect for solid-state cooling within a thermally insulated cabinet. An Arduino Uno is the common controller in this project, with a Pulse Width Modulation (PWM) algorithm being used to translate temperature differences into accurate control of the heating and cooling cycles for the improvement of power factor and temperature control performance relative to the traditional on/off control. An ESP8266 module is an IoT interface that communicates with the Arduino controller via the Inter-Integrated Circuit (I2C) bus for the coordination and coordination of real-time temperature data with the Blynk environment. Experimentally conducted results in an environment of 28°C showed that the prototype was capable of cooling a chamber to a temperature variation of 10-15°C in 45 minutes and maintaining it with a variation of ±0.5°C. Moreover, it could provide bi-directional synchronization with an end latency of less than one second through the IoT interface. In summary, this project aptly qualifies to be an effective implementation of a portable and steady-state solid-state refrigeration method for agricultural purposes. Future research would be based on optimizations of controls through Proportional-Integral-Derivative (PID) techniques and solar power.



| Advisor Signature | Group Leader Signature<br>(on behalf of the group) |
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# ABBREVIATIONS AND NOTATIONS

| | |
|---|---|
| ADC | Analog-to-Digital Converter |
| ASL | American Sign Language |
| EMG | Electromyography |
| I2C | Inter-Integrated Communication |
| MCU | Microcontroller Unit |
| SPI | Serial Peripheral Interface |
| UART | Universal Asynchronous Receiver / Transmitter |
| VR | Virtual Reality |

# ABSTRACT

Heat-sensitive products, such as agriculture seeds or pharmaceuticals, require precise and stable temperature control during their preservation in tropical climates like Vietnam. Traditional compressor-based refrigerators are usually bulky, require a high starting power input, and can transmit vibrations to sensitive items. The scope of this Capstone Project is to design and fabricate a "Smart Temperature Controlled Mini-Fridge" using a solid-state thermoelectric cooling technology-Peltier effect-with the integration of IoT monitoring. Results show that a prototype based on a Master-Slave architecture, where an Arduino Uno executes a PWM algorithm in order to dynamically adjust the cooling power via a MOSFET driver, is able to maintain target temperatures with high stability and seamlessly synchronize with IoT requirements, presenting itself as a compact, vibration-free solution for precision storage.

**Keywords:** Peltier Effect, Internet of Things (IoT), PWM Control, Smart Agriculture, Solid-State Refrigeration.

# CHAPTER I - INTRODUCTION

## 3.1. The Issues of Agriculture

Agriculture is the foundation of most countries worldwide. Although it has existed for a long time, agriculture continues to face numerous challenges.

### 3.1.1 Overseas agriculture

The agriculture sector in neighboring countries [1] is still grappling with inadequacies in agricultural disease monitoring, biofuel utilization, government assistance, and technical resources.

The US agricultural sector is facing numerous complex and multidimensional challenges that directly impact production, supply chains, and the industry's sustainable development. One of the most prominent issues is the highly pathogenic avian influenza (HPAI), which has caused severe damage to the livestock industry and food safety. The detection, monitoring, and control of this disease require closely coordinated measures and effective vaccination strategies to limit its spread and minimize impacts on agricultural trade.

Besides that, agricultural labor shortages have been consistently worsening, which negatively affects the whole value chain in agriculture. The result is reduced competitiveness of US agriculture in the global context and subsequently causes strain on local economies. The shortage of labor brings about urgent needs in the realms of labor policy reforms, expansion of the H-2A visa program for year-round agricultural workers and creating pathways for legalization of workers in the industry.

US agriculture faces challenges related to energy transition and biofuel development. The production and use of biofuels, such as renewable diesel, sustainable aviation fuel, and E15 ethanol, should increase if the goal of reducing dependency on fossil fuel is to be met with simultaneous enhancement of the rural economy and national energy security.

Another issue at hand is that there is a scarcity of skills and support facilities for farmers, especially young generation farmers. According to USDA statistics, new farmers (farmers with less than 10 years of farming experience) comprise 30% of total farmers, but 72% of these farmers are forced to have another source of income apart from agriculture as they have financial problems. It becomes imperative to provide technical support services and train them on managing business and farm transition to improve sustainability within the sector.

In a nutshell, the US agricultural sector faces a multitude of dilemmas and inhibiting factors, mainly in the control of diseases, labor issues, energy woes, as well as human resource development.

### 3.1.2 Agriculture in Vietnam

The agriculture sector in Vietnam [2] is still dominated by traditional production technologies with low technology complexity. The sector is characterized by resource intensive production with insignificant application of science, technology, and mechanics. The low performance contributes to poor regional and international competitiveness. Some sectors are far backward. The main reasons are small-scale production, low labor productivity, high cost of agricultural products, lack of linkage among sectors and within the sector itself, and several layers in the value chain.

The Vietnamese agriculture sector has always played an essential role in the socio-economic organism of the country and still remains the basis for ensuring the livelihood of rural residents and serves as the basis for the development of socio-economic and political stability to implement the aspirations for the industrialization and modernization of the country. In particular, during the period of the COVID-19 pandemic, while many sectors remained stagnant, agricultural production was still developing. By 2020, agriculture contributed about 14% to GDP and over 40% of employment.

*Figure 1: Severe drought and salinization lead to cracked fields, making farming increasingly difficult for local farmers.*

However, the agricultural sector has numerous serious challenges from external and internal perspectives. The issue of climate change poses the most critical challenge that has the potential to cause widespread flooding, drought, saltwater intrusion, and weather changes that have a direct impact on agricultural production, especially in the Mekong and Red River Deltas. The impact of drought and saltwater intrusion in Vietnam in 2020 reached hundreds of thousands of hectares, causing serious production and income losses for the agriculturists. The potential geopolitical changes and the possibility of readjustment in the international and regional supply chain also significantly pressurized the Vietnamese agricultural sector.

Internally, the Vietnamese agriculture industry is very extensive and labor-intensive and more focused on quantity and not quality. There are limited linkages between the industry and industry and service sectors for development. The overdependence on imported materials and technology as well as farm varieties and machinery and the development of export markets for the produce of a particular brand and origin make it difficult for Vietnamese agriculture to be competitive in the international market. High-tech crop varieties occupy just 30% of the total farm area.

The shortcomings lies in small-scale production, fragmentation of production, and low labor productivity. The average land per farm family is only 0.3 hectares, while the average agricultural labor productivity is only 1/10 of the country.

and then to developed countries in the region. The household farming model contributes 85% of overall production farm households, and has yet to be developed and keep up with trends in markets and Industry 4.0, lacking connectivity and cohesion in terms of how they produce and conduct business.

In short, agriculture in Vietnam faces severe challenges in terms of environment, economics, and production organization, and it needs proper policies to promote high technology in agricultural development, maximize quality and added value, and create value chain links in agricultural production.

## 3.2. Five significant issues in agriculture

### 3.2.1 Detecting pests to protect crops

Presently, the agricultural sector is facing an outbreak of disease cases on plants [3], where the major crops affected include dragon fruit and coconuts. In Long An province alone, there are more than 860 hectares of dragon fruit that have been damaged by the white spot disease, Binh Thuen province over 820 hectares, while Tien Giang province is also registering cases of the disease. Some farm owners have been compelled to harvest thousands of dragon fruit plants as the damaged fruit sells at 500 VND per kilogram, 20 times cheaper than fresh ones, resulting in losses that cost each farm hundreds of millions VND.

*Figure 2: Farmers have had to cut down thousands of diseased dragon fruit branches infected with white spot disease.*

In the case of coconut trees, three major harmful pests, including palm weevil, fruit borer, and coconut mites, are seriously damaging plantations, particularly in Ben Tre, Tra Vinh, and Tien Giang provinces. Palm weevils alone have damaged about 5,000 hectares of coconut trees, whereas fruit borers and coconut mites damaged almost 1,000 hectares of trees in each case. Sadly, a significant number of new plant species remain unnamed, and research work is stalled because of the constraints of the funding mechanism.

The quick dissemination of the disease, the evolution of new types of pests, as well as the constraints in research work, can be challenges to or have the potential to affect agriculture.

### 3.2.2 Consequences of abusing growth stimulants in cultivation

In recent years [4], the misuse of growth stimulants in agricultural cultivation has become an alarming trend in Vietnam. Various growth stimulant chemicals such as Gibberellin (GA), Auxin (NAA), and Cytokinin are being used under different trade names, especially for leafy vegetables such as water spinach, pumpkin, chayote, and celery. The importation of these chemicals has now been banned but most are being smuggled into the market due to their low cost and not being part of the approved list of the Ministry of Agriculture and Rural Resources.

*Figure 3: Emergency medical staff transporting a patient on a stretcher to the hospital*

The application of overdose without making a proper withdrawal period results in food poisoning cases time after time and may even result in cancer or death due to chemical residue accumulation in the body. In fact, vegetables can increase their biomass several times more with attractive appearance, but posing serious threat to public health, just 2-3 days after spraying growth stimulants. This not only has brought a grave threat to food safety, but also has shaken the confidence of consumers in domestic agricultural products.

**3.2.3 Minimize restrictions on transporting goods, import/export**

The agriculture sector is doing very well in Vietnam[5]. However, the logistics business has not really progressed very much. It can be noticed that with the export growth performance of over $53 billion for agriculture, forestry, and fishery products in 2023 and also with the overall record trade surplus at $13.06 billion, Vietnam's agriculture sector still has drastic logistics problems to deal with in transport pertaining to agri-products like durians from Dak Lak to Lang Son that require 7 days to export to China and with congestion tolls at 2.5 million VND per day per vehicle.

At present, the cost of logistics in terms of GDP is 20%, which is considerably higher compared to the international average of 11% of GDP. In terms of specific industries, the cost of logistics as a percentage of a product's cost is 12% for seafood products, 23% for wood products, 29% for vegetables and fruits, and 30% for rice. The loss rate of the agricultural supply chain is estimated to be between 25 and 30%, and for vegetables and fruits, up to 45%.

Moreover, a lack of large-scale, integrated production and transportation, and a reliance on foreign shipping companies, has made Vietnamese farm products less competitive in global markets.

Such logistics bottlenecks not only result in increased costs and longer timeframes for exports but also have a direct effect on the quality and added value of agricultural products. This therefore gives room for the need to optimize the logistics system within the agricultural chain.

#### 3.2.4 Find the most optimal environment for plants, limiting dependence on weather

The crop production sector is facing big challenges in Binh Thuan due to abnormal weather and increased input costs[6]. The total cultivated area for short-term crops in 2022 was 193,400 hectares, attaining 96.41% compared to the same period, but the rice cultivation area achieved only 92.52% of the plan, mainly due to the sharp price increases of fertilizers and plant protection chemicals, which forced many farmers to give up their crops. This was largely contributed by unprecedented continuous rainfall causing local flooding, forcing residents in some areas to stop production and summer-autumn crop cultivation to attain only 95.85% of the plan.

*Figure 4: Rice harvesting using a combination harvester in a paddy field.*

In addition, dragon fruits have had continuously dropping prices to merely 2,000-10,000 VND per kg with rising production costs and reduced consumption at Chinese border gates, causing many orchard farmers to abandon their gardens and grow another type of crop. Moreover, pest and disease strains such as brown planthoppers on rice, brown spot diseases on dragon fruits, mosaic diseases on cassava, and fall armyworm on corn are still developing complicatedly with risks of outbreak from being epidemics.

All the above-mentioned issues prove that the crop production industry in the province of Binh Thuan is significantly impacted by climate change, the increase of prices for production materials, as well as the consumption markets. There is a great need for response solutions in order for the industry to be sustainable.

### 3.2.5 Subsidized rice seeds are moldy

Most agricultural produce in Vietnam is grown under a tropical monsoon climate with high moisture content and temperature fluctuations [7]. These conditions are particularly unkind to post-harvest preservation. Seeds are highly hygroscopic materials; when exposed to poor environmental conditions, they rapidly absorb moisture and thus create a suitable environment for fungal growth and metabolic acceleration. This then leads to "seed aging," in which the germination rate suddenly falls before planting starts. Though seed quality is such a crucial factor in crop yield, so many regions lack proper storage facilities in order to shield these biological assets from the harsh local climate.

The seriousness of this problem has been brought to light by a report from Dan Viet Newspaper , citing a mass case in Chau Thanh District, Tra Vinh Province, where a lot of quality subsidized rice seeds handed to famers as a form of government aid in a support program had seriously deteriorated as they reached the recipients. In this case, the report cited that upon opening the bags of seeds, the grains appeared to be affected by mold, unfitting color, and an unpleasant smell due to accumulated moisture. This made the germination rate of the seeds, despite an effort to plant them, almost nothing, an indication that the seeds' embryos had died because of a poor preservation environment that might have been encountered before planting.

The failure to maintain proper storage conditions resulted in immediate economic consequences for the farming households. Not only were the subsidized resources wasted, but farmers were also forced to incur additional costs to purchase replacement seeds from the market to meet the seasonal schedule. This incident underscores a critical gap in the agricultural supply chain: the lack of accessible, temperature-controlled storage solutions. It demonstrates that without active intervention to control temperature and humidity—such as the solution proposed in this project—high-value agricultural products remain vulnerable to spoilage, leading to financial instability for farmers.

*Figure 5: The rice was damaged due to mold.*

### 3.3. Aim of the research

This research therefore aims to design and implement a Smart Thermoelectric Refrigeration System integrated with IoT technology. The project basically tries to provide a compact, vibration-free, and cost-effective cooling solution capable of maintaining precise temperature ranges of 10−25 ∘ C. This will address the limitations on conventional compressor-based refrigerators by conceptualizing stability and remote manageability in a system.

With this overall aim, the research objectives are directed at achieving these particular goals:

The solid-state cooling will be implemented by using Peltier technology, TEC1-12706, as the primary means of cooling. It does not require refrigerant gases or compressors, hence an eco-friendly and zero-vibration system.

Sophisticated Control Algorithm: The design of an Advanced Control Algorithm for Pulse Width Modulation (PWM) control logic based on the Arduino platform. The algorithm adjusts the cooling rate based on the error value between the current and set temperatures. The algorithm is an improvement over the On/OFF control algorithm.

IoT Based Remote Monitoring System: To design an efficient remote monitoring system by utilizing the ESP8266 Wi-Fi module and Blynk platform, such that it enables the user to monitor the readings as well as control the system from any distant place through the smartphone.

System Integration: The task involves combining all subsystems related to mechanical, electrical, and software components and packaging them; then, it is required to verify their heat dissipation capabilities under normal room conditions as it is housed inside a thermally insulated box.

# CHAPTER II - DESIGN SPECIFICATIONS AND STANDARDS

In this chapter, the technical details related to the proposed system, the standards used in the design process, and the feasible constraints have been explained.

## 4.1. Design specifications

The design of a system involves hardware and software environments.

### 4.1.1 Hardware

The physical system is designed with the use of modular electronic components that are integrated into a thermally insulated cabinet. The important hardware requirements are as listed in Table below.

*Table 1: The Hardware of Design Specifications.*

| No. | Component | Parameter / Specification |
|---|---|---|
| 1 | **Main Controller** | **Arduino Uno R3** (Microcontroller: ATmega328P; Operating Voltage: 5V; Clock Speed: 16 MHz) |
| 2 | **IoT Gateway** | **ESP8266 NodeMCU** (Wi-Fi Standard: 802.11 b/g/n; Operating Voltage: 3.3V; Communication: I2C Master) |
| 3 | **Cooling Engine** | **Peltier Module TEC1-12706** (Vmax: 15V; Imax: 6A; Qmax: ~60W) |
| 4 | **Power Driver** | **MOSFET IRF540N** (Vdss: 100V; Id: 33A; Type: N-Channel Low-side switching) |
| 5 | **Temperature Sensor** | **DS18B20 Waterproof** (Range: -55°C to +125°C; Accuracy: ±0.5°C; Interface: 1-Wire Digital) |
| 6 | **Display** | **LCD 16x2 with I2C Module** (Address: 0x27; Backlight: Blue) |

| | | |
|---|---|---|
| 7 | **Power Supply** | DC source: 12V DC; Current Capacity: 30A (360W); |

### 4.1.2 Software

The control logic and user interface are developed using standard embedded programming tools and IoT platforms, as shown in Table below.

*Table 2: The Software of Design Specifications.*

| No. | Component | Specification |
|---|---|---|
| 1 | **Programming Language** | **C++** (Embedded C for Arduino/ESP) |
| 2 | **Development Environment** | **Arduino IDE 2.0+** (Libraries: DallasTemperature, Wire.h, BlynkSimpleEsp8266) |
| 3 | **IoT Platform** | **Blynk IoT Cloud** (Protocol: Virtual Pins V0-V4; Interface: Mobile App & Web Dashboard) |
| 4 | **Communication Protocol** | **I2C Inter-Integrated Circuit** (Speed: 100 kHz Standard Mode; Master: ESP8266 / Slave: Arduino Addr 0x08) |

## 4.2. Engineering Codes and Standards

For ensuring safety, reliability, and compatibility, the following international engineering standards are being followed in the project:

*Table 3: Engineering Codes and Corresponding Standards.*

| Element/System | Engineering Standards | Comments |
|---|---|---|
| **Wi-Fi Connectivity** | **IEEE 802.11** | Standard for Wireless LAN communication used by the ESP8266 module to connect to the Blynk server. |
| **Temperature Sensors** | **IP68 Rating** (IEC 60529) | The DS18B20 probe is certified IP68 (Dust tight and immersion beyond |

| | | |
|---|---|---|
| | | 1m), ensuring durability inside the humid fridge environment. |
| **Electrical Enclosure** | **NEMA Type 1 / IP40** | The SINO electrical box provides protection against solid objects and incidental contact with enclosed equipment (Safety for users). |
| **Programming** | **ISO/IEC 14882** | Programming language standard for C++, ensuring the firmware is structured and maintainable. |
| **Data Communication** | **I2C Specification** (NXP UM10204) | Adheres to the standard protocol for 2-wire serial communication between microcontrollers. |
| **Refrigeration Safety** | **RoHS Compliance** | The Peltier modules and electronic components are selected to be free of hazardous substances (Lead-free), making them safe for agricultural use. |

## 4.3. Realistic Constraints

The project design and development activities were impacted by realistic constraints listed in Table below.

*Table 4: Realistic Constraints.*

| Area | Realistic Constraints |
|---|---|
| **Economic** | **Budget Limitation:** As a student project, the total cost was capped at 2,000,000 VND (~$80). This necessitated the use of cost-effective components like the Arduino Uno and Peltier modules |

| | |
|---|---|
| | instead of expensive industrial PLCs or compressors. |
| **Environmental** | **Energy Efficiency:** The Peltier module has lower thermal efficiency (COP) compared to gas compressors. To mitigate this, a PWM control algorithm was implemented to reduce power consumption when the target temperature is reached. |
| **Health and Safety** | **Electrical Safety:** The system operates at 12V DC (Safe Extra Low Voltage). However, the AC/DC adapter connects to 220V mains. All high-voltage connections are enclosed within the SINO box to prevent electric shock. |
| **Manufacturability** | **DIY Fabrication:** The fridge chassis was modified from a Styrofoam insulation box manually. The precision of thermal sealing relies on hand-craftsmanship rather than industrial molding, which may affect cooling retention slightly. |
| **Social** | **User Accessibility:** The system is designed for farmers who may not be tech-savvy. Therefore, the interface includes both physical buttons (simple interaction) and a mobile app (advanced monitoring), ensuring ease of use for all demographics. |

# CHAPTER III - PROJECT MANAGEMENT

The Project Management Plan describes the manner in which the “Smart Mini-Fridge” project will be carried out, tracked, and controlled. Additionally, it provides a budget for financials, an activity timeline, as well as human resource allocation in an efficient manner in which all the requirements are attained.

## 5.1. Budget and Cost Management Plan

The project must work within a tight budget constraint imposed on it as a student project. The cost estimate of some of the major constituents and materials used is presented in Table below. The total cost estimate is roughly 2,000,000 VND, which is equivalent to around 80 USD.

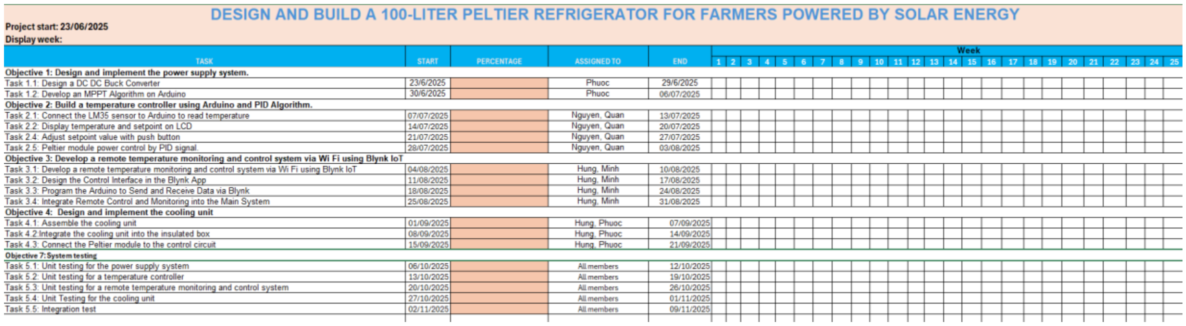
Table 5.0: Price for components.

| Item | Description | Quantity | Unit Price (VND) | Total Cost (VND) |
|---|---|---|---|---|
| **Microcontroller** | Arduino Uno R3 + Cable | 1 | 150,000 | 150,000 |
| **IoT Module** | ESP8266 NodeMCU | 1 | 90,000 | 90,000 |
| **Cooling Engine** | Peltier TEC1-12706 | 2 | 45,000 | 90,000 |
| **Dissipation System** | Heatsink + High RPM Fans (Sunon) | 1 set | 250,000 | 250,000 |
| **Power Driver** | MOSFET IRF540N Module | 2 | 70,000 | 35,000 |
| **Sensors & Display** | DS18B20 + LCD 16x2 I2C | 1 set | 100,000 | 100,000 |
| **Power Supply** | Adapter 12V-15A | 1 | 350,000 | 350,000 |

| | | | | |
|---|---|---|---|---|
| | (Industrial) | | | |
| **Chassis & Case** | Styrofoam Box + SINO Enclosure | 1 set | 200,000 | 200,000 |
| **Miscellaneous** | Wires, Breadboard, Glue, Screws | - | 200,000 | 200,000 |
| **Contingency** | Risk buffer (10%) | - | - | 150,000 |
| **Total Estimated Cost** | | | | 1,096,000 |

## 5.2. Project Schedule

In this part, show the project schedule when implementation as shown in Figure below.



DESIGN AND BUILD A 100-LITER PELTIER REFRIGERATOR FOR FARMERS POWERED BY SOLAR ENERGY

Project start: 23/06/2025

Display week:

| TASK | START | PERCENTAGE | ASSIGNED TO | END |
|---|---|---|---|---|
| **Objective 1: Design and implement the power supply system.** | | | | |
| Task 1.1: Design a DC DC Buck Converter | 23/6/2025 | | Phuoc | 29/6/2025 |
| Task 1.2: Develop an MPPT Algorithm on Arduino | 30/6/2025 | | Phuoc | 06/07/2025 |
| **Objective 2: Build a temperature controller using Arduino and PID Algorithm.** | | | | |
| Task 2.1: Connect the LM35 sensor to Arduino to read temperature | 07/07/2025 | | Nguyen, Quan | 13/07/2025 |
| Task 2.2: Display temperature and setpoint on LCD | 14/07/2025 | | Nguyen, Quan | 20/07/2025 |
| Task 2.4: Adjust setpoint value with push button | 21/07/2025 | | Nguyen, Quan | 27/07/2025 |
| Task 2.5: Peltier module power control by PID signal. | 28/07/2025 | | Nguyen, Quan | 03/08/2025 |
| **Objective 3: Develop a remote temperature monitoring and control system via Wi Fi using Blynk IoT** | | | | |
| Task 3.1: Develop a remote temperature monitoring and control system via Wi Fi using Blynk IoT | 04/08/2025 | | Hung, Minh | 10/08/2025 |
| Task 3.2: Design the Control Interface in the Blynk App | 11/08/2025 | | Hung, Minh | 17/08/2025 |
| Task 3.3: Program the Arduino to Send and Receive Data via Blynk | 18/08/2025 | | Hung, Minh | 24/08/2025 |
| Task 3.4: Integrate Remote Control and Monitoring into the Main System | 25/08/2025 | | Hung, Minh | 31/08/2025 |
| **Objective 4: Design and implement the cooling unit** | | | | |
| Task 4.1: Assemble the cooling unit | 01/09/2025 | | Hung, Phuoc | 07/09/2025 |
| Task 4.2:Integrate the cooling unit into the insulated box | 08/09/2025 | | Hung, Phuoc | 14/09/2025 |
| Task 4.3: Connect the Peltier module to the control circuit | 15/09/2025 | | Hung, Phuoc | 21/09/2025 |
| **Objective 7: System testing** | | | | |
| Task 5.1: Unit testing for the power supply system | 06/10/2025 | | All members | 12/10/2025 |
| Task 5.2: Unit testing for a temperature controller | 13/10/2025 | | All members | 19/10/2025 |
| Task 5.3: Unit testing for a remote temperature monitoring and control system | 20/10/2025 | | All members | 26/10/2025 |
| Task 5.4: Unit Testing for the cooling unit | 27/10/2025 | | All members | 01/11/2025 |
| Task 5.5: Integration test | 02/11/2025 | | All members | 09/11/2025 |

| Week | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

*Figure 6: Gantt Chart.*

## 5.3. Resource Planning

To maximize efficiency, the team structure is organized based on individual strengths. The roles and responsibilities are defined as follows:

| Member Name | Role | Responsibilities |
|---|---|---|
| Do Buu Phuoc | Project Manager & System Architect | • **Ideation:** Conceptualize the original idea and system architecture. **Management:** Assign tasks, monitor progress, and manage the budget. **Troubleshooting:** Provide technical support and solve critical issues across all objectives. **Documentation:** Compile and write the final Capstone Report. |
| Nguyễn Thế Hưng | Mechanical Engineer | • **Responsible for Objective 1:** Cooling Unit Implementation. Assemble the Peltier modules, heatsinks, and fans. Modify the fridge chassis and ensure thermal insulation. Install the physical components into the housing. |
| Nguyễn Minh Quân | Embedded Systems Engineer | • **Responsible for Objective 2:** Controller & Logic. Design the circuit diagram and wiring on the breadboard. Program the Arduino Uno with the PWM control algorithm. Interface the sensors and MOSFET driver with the microcontroller. |

| | | |
|---|---|---|
| Văn Thảo Minh | IoT Software Engineer | • **Responsible for Objective 3:** IoT Monitoring System. Develop the firmware for ESP8266 to communicate via I2C. Design the UI/UX on the Blynk Mobile App and Web Dashboard. Ensure data synchronization between the device and the cloud. |
| Nguyễn Hồng Hải Nguyên | **Support Engineer & Presenter** | • **Support for Objective 3:** Assist in testing Wi-Fi connectivity and debugging the IoT interface. **Presentation Lead:** Prepare the slide deck and represent the team during the final defense presentation. Assist the Project Manager in documentation review. |

# CHAPTER IV - LITERATURE REVIEW

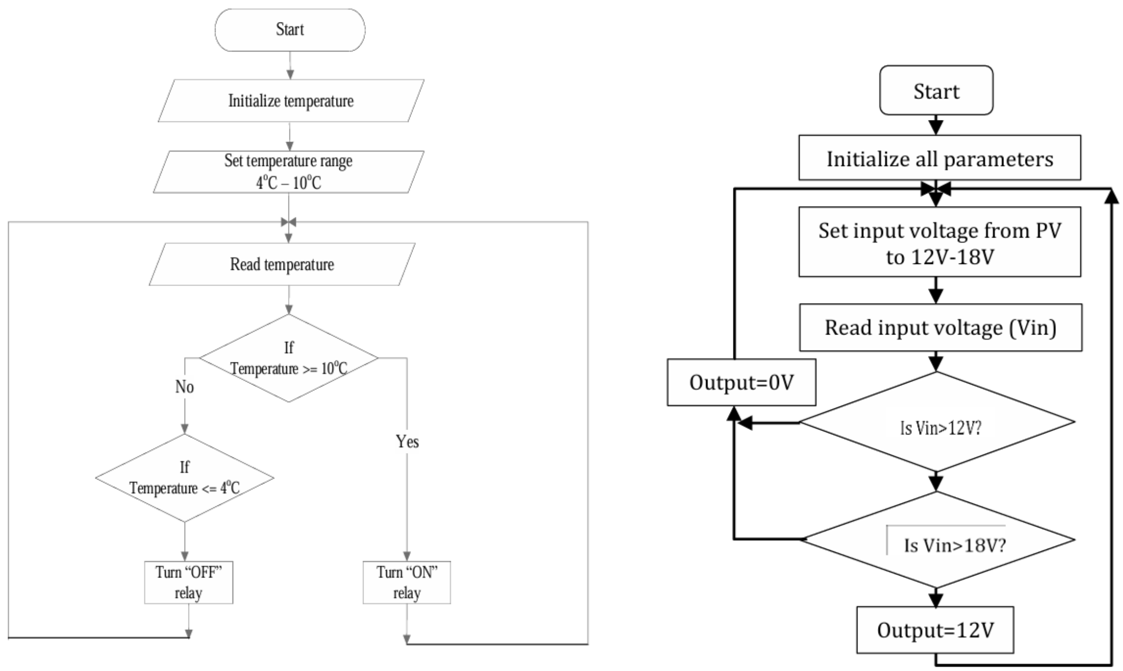
This chapter looks into the literature on important research works and available technical solutions on cold storage systems for agricultural products, thermoelectric cooling devices, and IoT-related monitoring systems. This literature review seeks to provide analysis on the strengths and weaknesses of past research works and will help create a theoretical basis for developing the Smart Temperature Controlled Mini-Fridge.

## 6.1. Cold Storage Solutions for Agriculture

The preservation of quality in such commodities like fruits, vegetables, and seeds in tropical regions poses a very important challenge.

Agbo David Odu et al. (2024) [8] reported on the design and development of a DC - powered fridge intended for storing different vegetables and fruits. Their proposed concept used a DC compressor and PV system with a battery to provide a storage temperature that ranged from $4\,{\circ}\,C$ to $10\,{\circ}\,C$. It was pointed out by Agbo et al. that incorporating renewable energy into cold storage can result in considerable post-harvest loss reduction.

**Relevance to this project**: This verifies that active cooling is a very important requirement in farming. Despite this, because this system uses a DC compressor, it is bulkier, heavier, and prone to vibrations, which may not be appropriate for storing biological samples such as seeds, which are supposed to be in vibration-free environments.




(a): Temperature Control Flow Chart

(b): Buck Converter Flow Chart

*Figure 7: Flowchart of the DC Fridge*

## 6.2. Thermoelectric Cooling Technology (Peltier Effect)

Compressor-based cooling systems had certain shortcomings (large size, vibration, refrigerant use), which sparked interest in more modern research into cooling by Peltier element cooling.

Prem Ritesh Kumar et al. (2019) [2] developed a "Portable Thermoelectric Refrigerator" based on the TEC1-12706 thermoelectric cooler module. The study revealed that the Peltier refrigerator provides some unique features such as its eco-friendly operation without the use of any CFCs, compactness, portability, and vibration-less performance. Their model worked successfully to lower the temperature inside from 36°C to 18°C within one hour with the Cop (COP) of about 1.1 (theoretical).

**Critical Finding**: This proves the practicality of the use of Peltier modules in portable coolers in which the temperature range of 15-20°C is enough to preserve insulin or fresh produce.

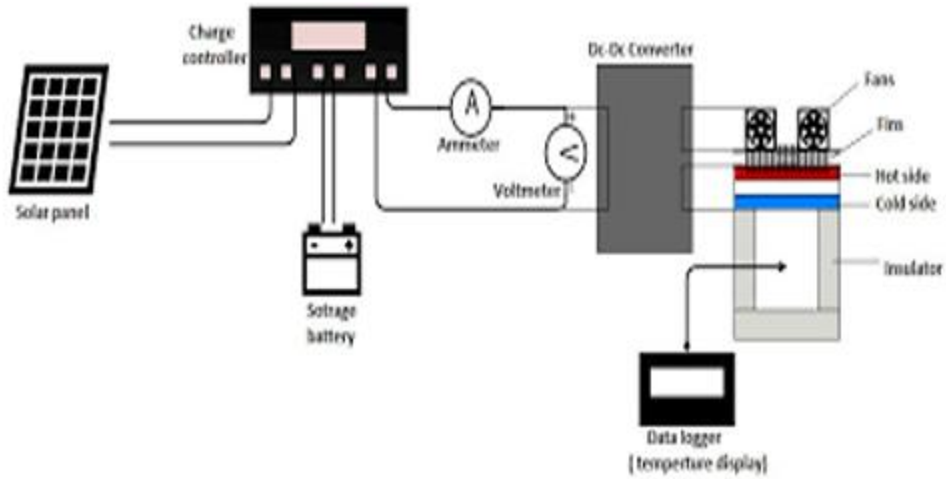



*Figure 8: Connection.*

In addition to that, Muhammad Fairuz Remeli et al. [3] in the year 2020 carried out the project "Experimental study of mini cooler by Peltier thermoelectric cell." In fact, they highlighted the process describing the thermal resistance model in the heat dissipation part. Further results validated that the Peltier module alone can take away 25W of power with the help of appropriate heat dissipation. This helped to cool the interior to 18.5°C with the COP variation of 0.52-0.53.

**Important Finding**: The important point made in this study concerns the heat dissipation system in Peltier modules. The authors feel that the COP value, although lower than that in the compression-type module, is sufficient for small volume cooling if the heat on the hot side can be taken care of.

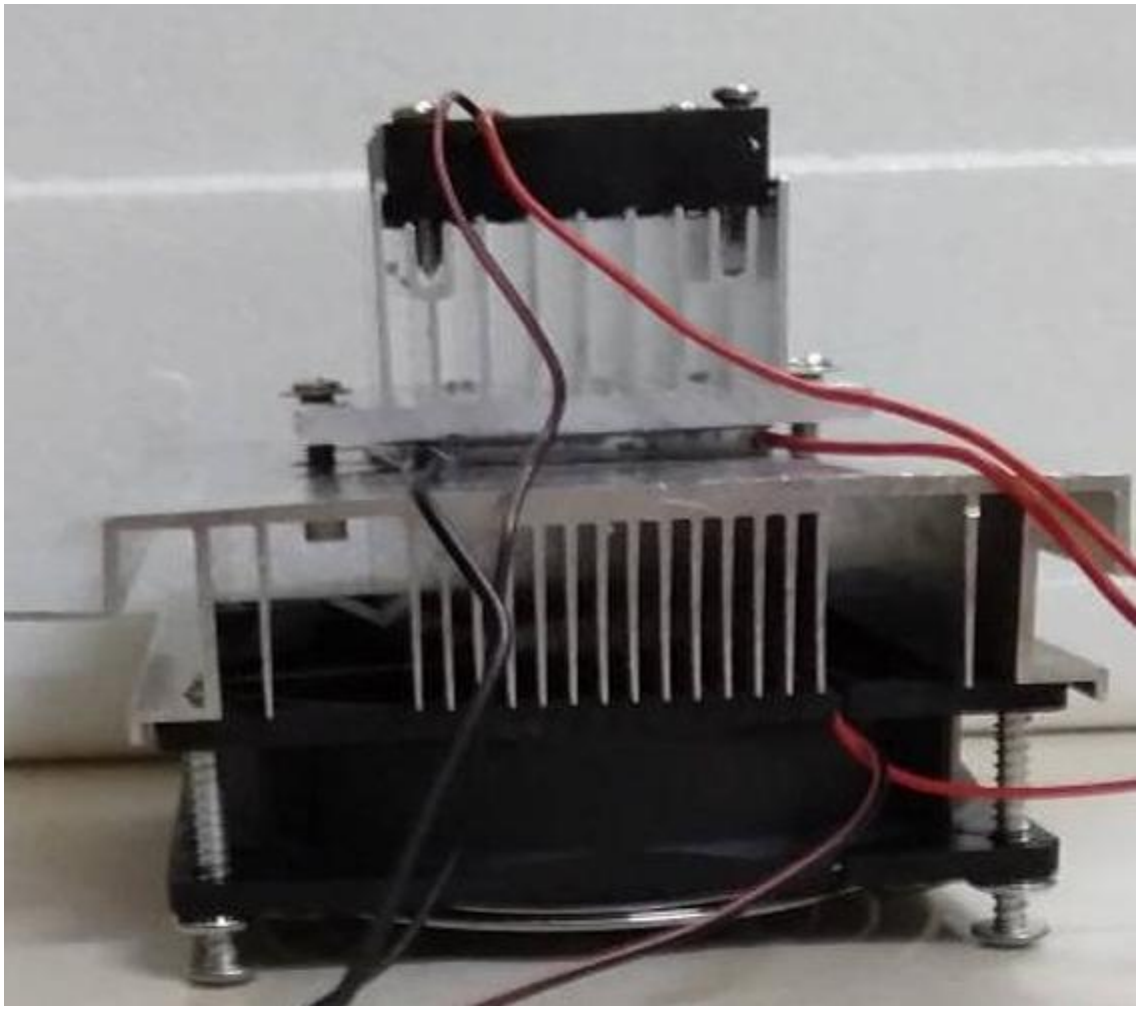


*Figure 9: Installation of the Peltier module with the heat sinks and cooler fans.*

### 6.3. IoT-Based Monitoring and Control

Within modern agriculture, temperature must be controlled on a constant basis; otherwise, produce can spoil due to a breakdown in equipment.

A paper titled "IoT Based Temperature Monitoring System Using ESP8266 & Blynk App" was proposed by Jella Bharath Kumar et al. in 2024. The authors' proposal aimed to use a NodeMCU ESP8266 module combined with a DS18B20 temperature sensor and an OLED display to remotely track temperature readings using the Blynk cloud service. The proposed work enabled users to display real-time readings from a smartphone device and enabled them to establish a temperature limit to turn relays (Heater/Cooler) ON/OFF.

**Key Insight**: This study verifies the feasibility and viability of the ESP8266 + Blynk system design in implementing far-off environmental monitoring. It lays down the technology groundwork in the implementation of the 'Smart' feature in our Mini Fridge Project

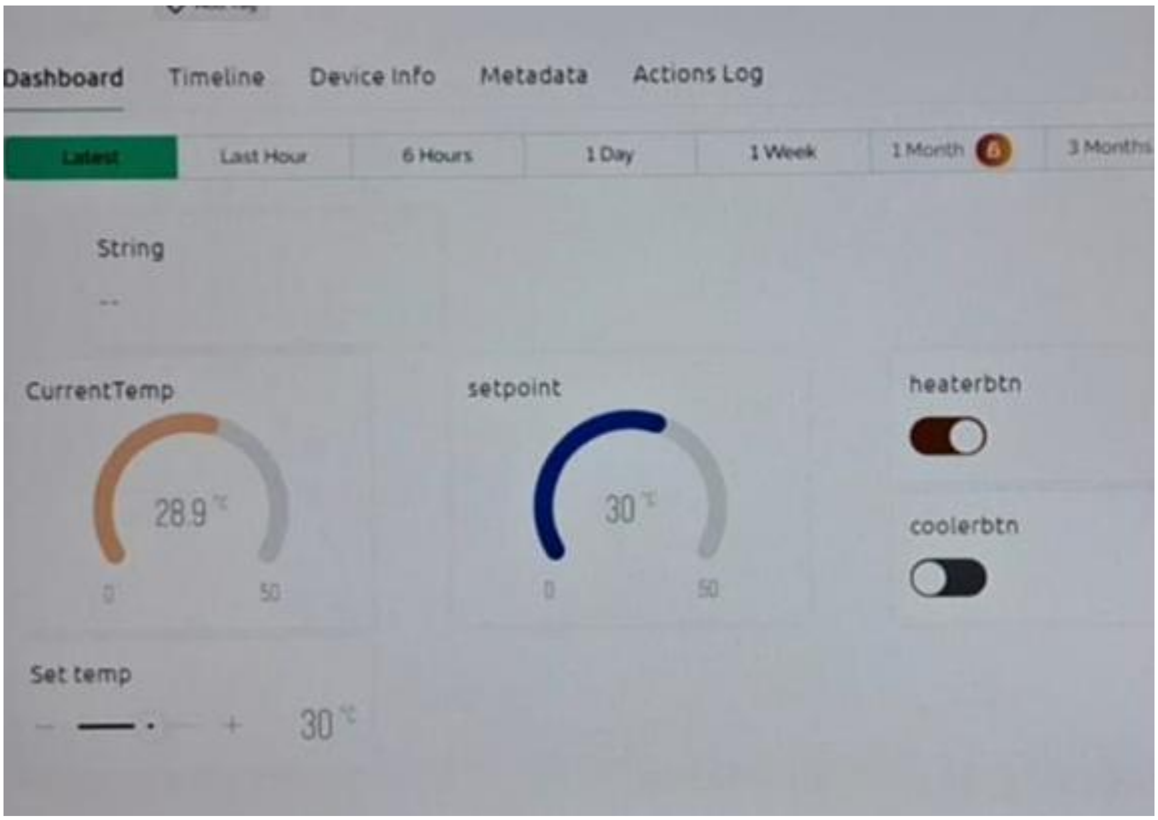



*Figure 10: Blynk App Interface and output.*

## 6.4. Summary and Research Gap

Several conclusions and gaps based on the reviewed literature are identified as follows:

1. Compressor systems Agbo et al. are potent but too bulky and cause vibration, unsuitable for delicate seeds.
2. Most existing Peltier refrigerators, as explained by Prem Ritesh Kumar et al., are controlled by simple ON/OFF methods or are directly connected to radiation,

which causes temperature fluctuation and allows them to achieve a lower energy efficiency.

3. The studies in IoT normally revolve around monitoring by Jella Bharath Kumar et al., rather than focus on the control logic.

**Conclusion**: Solutions that integrate the three aspects mentioned, namely vibration-free Peltier cooling, logic for precise PWM control that would stabilize temperature and thus save energy, and IoT connectivity, all especially targeted at small agriculture needs-like seed germination or storage of high-value produces, among others-are lacking. This Capstone project tries to fill this lacuna by working on a Smart Thermoelectric Mini-Fridge integrating these three into one compact, small-size, and low-cost device.

# CHAPTER V – METHODOLOGY

This chapter outlines the step-by-step process used to create and build the Smart Temperature Controlled Mini-Fridge. The approach follows the goals set out in the system design spelling out the steps needed to move from the drawing board to the finished prototype.

## 7.1. Goal and System Overview

### 7.1.1 Project Goal

The primary goal is to transform a standard insulated container into an active cooling refrigerator using solid-state thermoelectric technology (Peltier). The system is designed to maintain precise temperature ranges ( 10 – 25 C 10−25 C ) using a PWM-based control algorithm running on an Arduino platform. Additionally, the project aims to implement a comprehensive IoT monitoring system via ESP8266 and Blynk, allowing users to monitor temperature and switch operating modes remotely.

### 7.1.2 System Overview

The system operates based on a Closed-Loop Control architecture integrated with an IoT gateway. The process flow is defined as follows:

1. **Input**: Temperature data is acquired from the DS18B20 sensor and user commands are received from either physical buttons or the Blynk mobile app.

2. **Process**: The Arduino Uno (Slave) processes the thermal data and calculates the required cooling power using PWM logic. Simultaneously, the ESP8266 (Master) manages Wi-Fi connectivity and synchronizes data with the cloud.

3. **Output**: The system drives the Cooling Unit (Peltier + Fans) via a MOSFET driver and updates the status on both the local LCD and the remote Dashboard.

## 7.2. Implementation Objectives and Tasks

### 7.2.1 Objective 1: Cooling Unit Implementation

This chapter outlines the step-by-step process used to create and build the Smart Temperature Controlled Mini-Fridge. The approach follows the goals set out in the system design spelling out the steps needed to move from the drawing board to the finished prototype.

#### 7.2.1.1. Task 1.1 - Thermoelectric Module Assembly

The construction of the core cooling engine involves putting together the thermoelectric modules (TEC1-12706). To make sure heat transfers well, we spread a thin even layer of high-quality thermal compound (heatsink paste) on both ceramic sides of the Peltier modules. We then place these modules between big aluminum finned heatsinks on the hot side and solid aluminum cooling blocks on the cold side. We tighten this setup with stainless steel screws to apply just the right pressure, which helps to lower the thermal contact resistance between the layers.

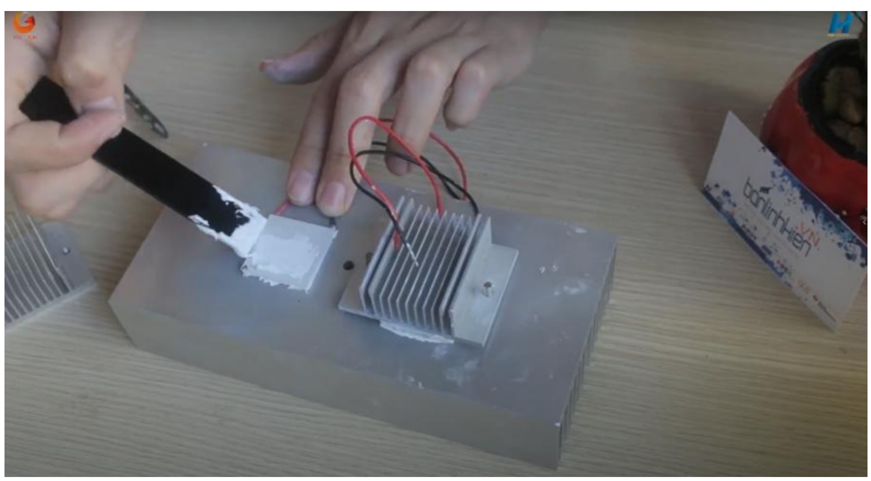


*Figure 11: Assembling process of the cooling unit with thermal paste.*

#### 7.2.1.2. Task 1.2 - Airflow and Dissipation Management

As the efficiency of a Peltier cooler largely relies on the existing temperature difference ( $\Delta T$ ) between the two sides of the cooler, active refrigeration becomes a must in our system design. High-speed DC fans (Sunon) are arranged on the outer heat sinks to effectively blow out the produced heat to the environment. At the same time, lower-speed fans are placed on the internal cold sinks to strongly circulate the cooled air inside the insulated box. This process creates a uniform temperature field inside the fridge due to convection currents. Furthermore, it prevents any frost buildup on the cold sinks.


*Figure 12:Heatsink fan assembly.*

#### 7.2.1.3. Task 1.3 - Chassis Integration and Insulation

The refrigerator chassis acts as the thermal envelope. Modification, such as making precise rectangular cut-outs on the side wall of the chassis for the unit, is to be done. When inserted, the mechanical assembly-chassis interface is sealed with industrial-grade insulation foam and silicone sealant. This step is critical to avoid "thermal bridging" or leakage of heat from outside and to ensure that the modification retains structural integrity.

*Figure 13: DIY peltier fridge insulation.*

### 7.2.2 Objective 2: Build a temperature controller using Arduino and PWM Control Logic

This objective entails the creation and development of the Central Processing Unit whose mandate would be to read environmental information as well as execute control algorithms.

#### 7.2.2.1. Task 2.1 - Control Circuit Design and Prototyping

The first step in the development of the control circuit would be to prototype it on a breadboard and test the electrical connections. This would entail connecting the Arduino Uno circuit board and the DS18B20 digital temperature sensor together by means of a 4.7kΩ pull-up resistor in order to enable 1-Wire communication. At the same time, the connection of the user interface components such as the I2C LCD 16x2 module and push-button switches would be done with debounce capacitors.

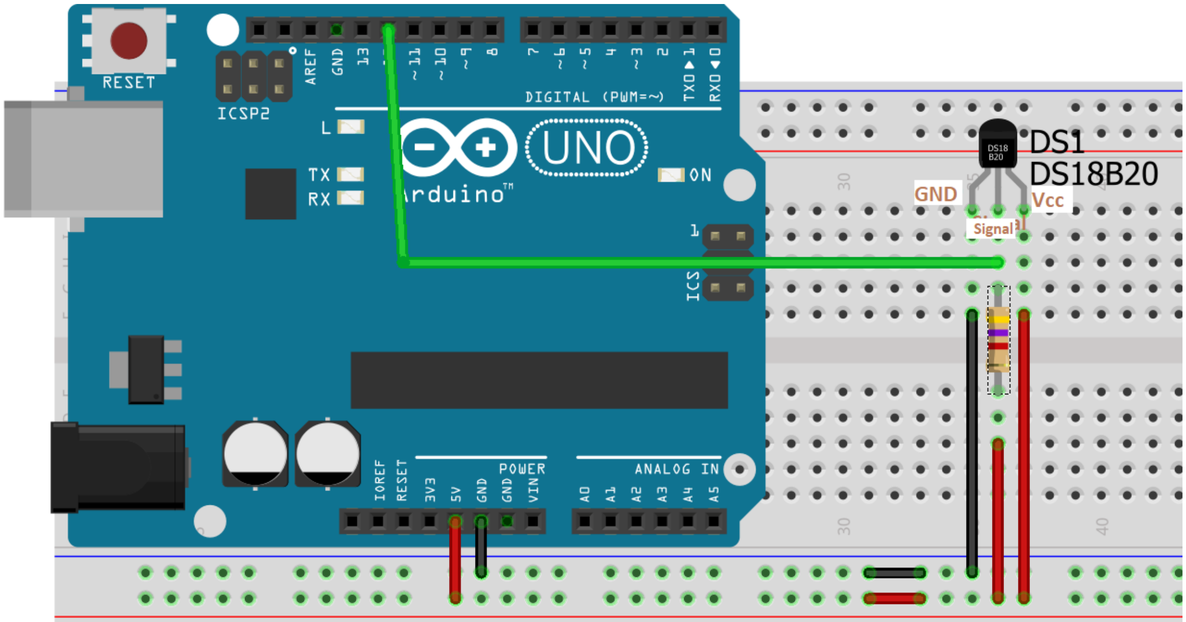



*Figure 14: Arduino UNO DS18B20 wiring diagram.*

#### 7.2.2.2. Task 2.2 - Power Driver Interface Implementation

Because the Arduino's GPIO is 5V/40mA, it cannot drive the high-powered Peltier elements directly (12V/6A). Hence, an IRF540N MOSFET module is incorporated to serve as an electronically controlled high-speed switch. The PWM output pin of the Arduino is wired to the Gate of the MOSFET. In this way, the microcontroller is capable of controlling the 12V power source to drive the cooling subsystem.

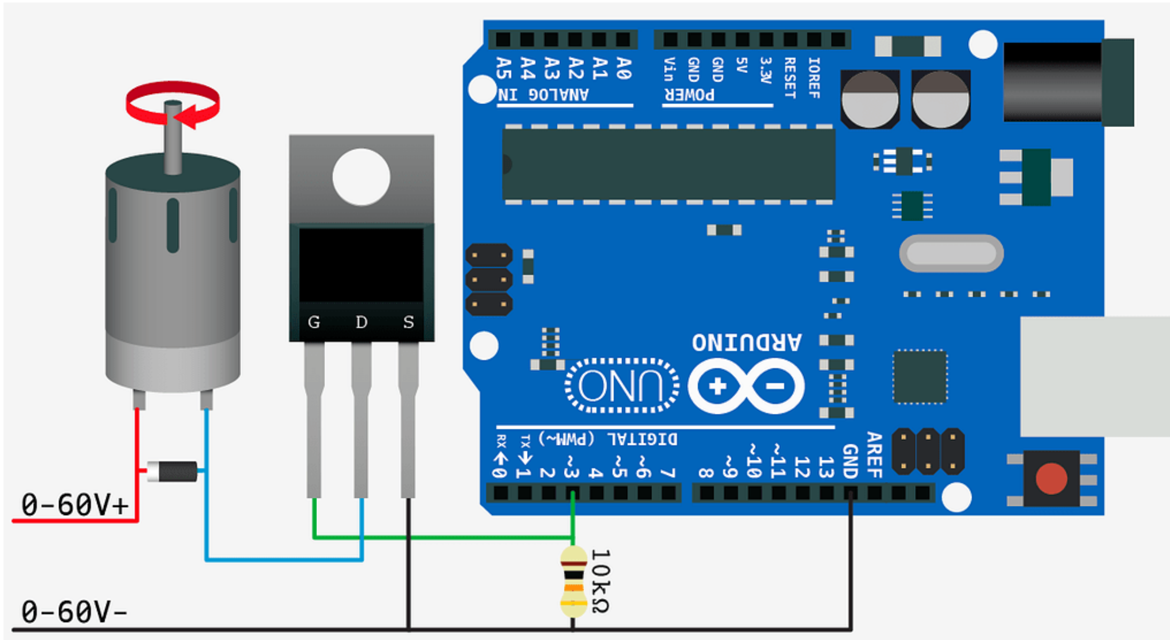



*Figure 15: Mosfet module Arduino.*

#### 7.2.2.3. Task 2.3 - Embedded Firmware Development

The firmware of this system is written in C++ with Arduino IDE. The main purpose of this firmware is to develop PWM Control Logic. The system is an advancement from a simple thermostat, which turns on and offs power, and this firmware calculates the error between the PV and SP, which are the current and set temperatures, respectively, and maps this error into a duty cycle, which is 0-255, and thus lowers the speed of fans and Peltier elements when temperatures are approaching target values and thus increases energy efficiency.


*Figure 16: PWM duty cycle graph.*

### 7.2.3 Objective 3: IoT Monitoring System

This objective is concerned with the development of the connectivity layer, which will enable the device to communicate with the internet.

#### 7.2.3.1. Task 3.1 - Master-Slave I2C Communication

First, we set up an independent Master-Slave architecture to isolate the critical tasks of control from network operations. The ESP8266 is the Master and is connected with Arduino, which works as a Slave. It uses the I2C protocol for the connection through SDA and SCL pins. We program this firmware in such a way that the ESP8266 will periodically query the Arduino for the present temperature and operating mode every 300ms. Meanwhile, it might start sending asynchronous "Write" commands to the Slave when any user on the mobile application triggers an action.

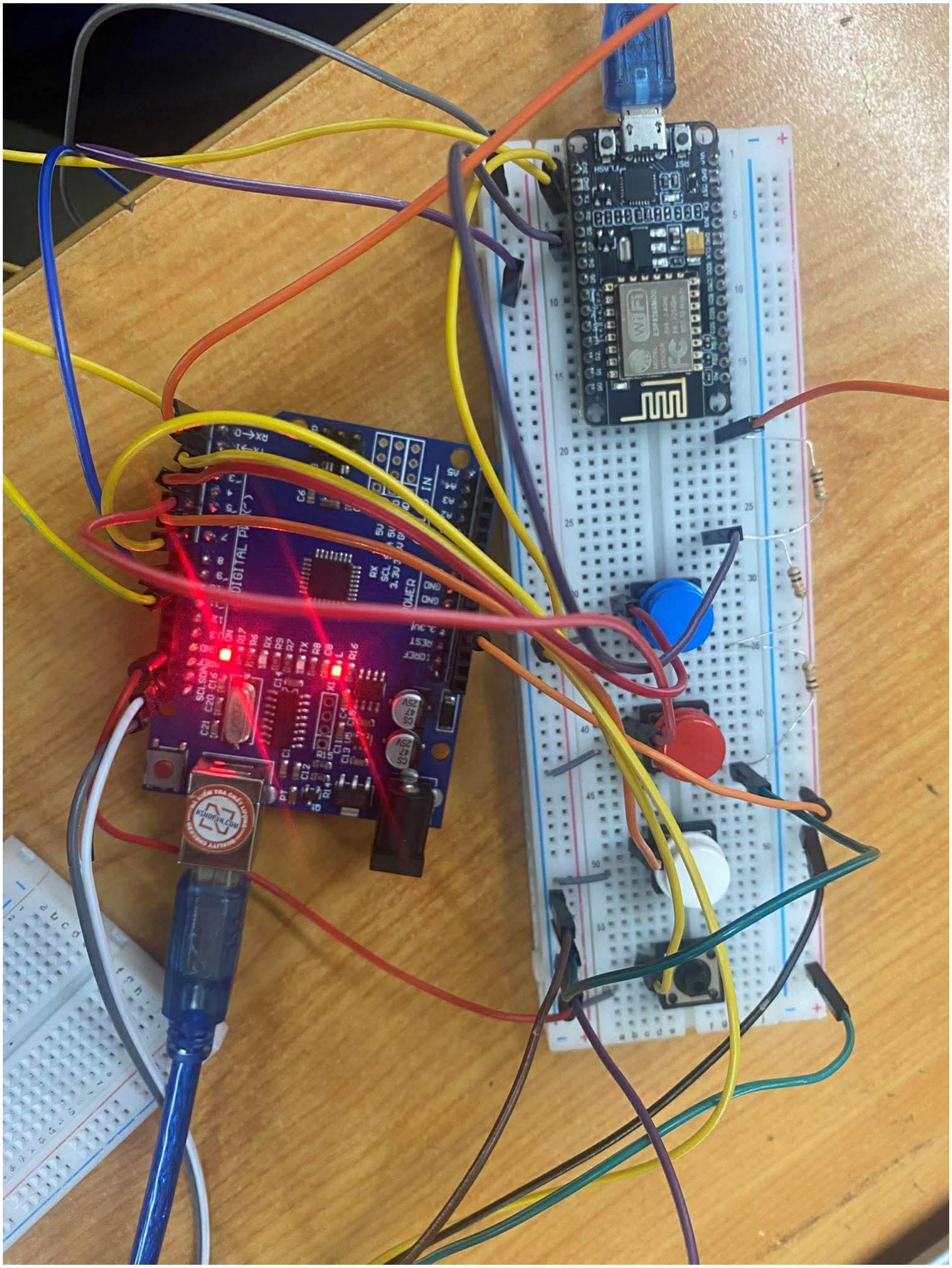


*Figure 17: Arduino ESP8266 I2C connection diagram.*

#### 7.2.3.2. Task 3.2 - Blynk Dashboard Configuration

The remote user interface is developed using the Blynk IoT Platform. A type of "Gauge" widget is set up where real-time temperature readings (measured from 0 to 50˚C) are obtained from the temperature sensors. Moreover, four separate "Switch" widgets are dedicated to Virtual Pins (V1, V2, V3, and V4) to switch ON/OFF or choose the defined temperature ranges of either 10-15 C, 15-20 C, or 20-25 C.

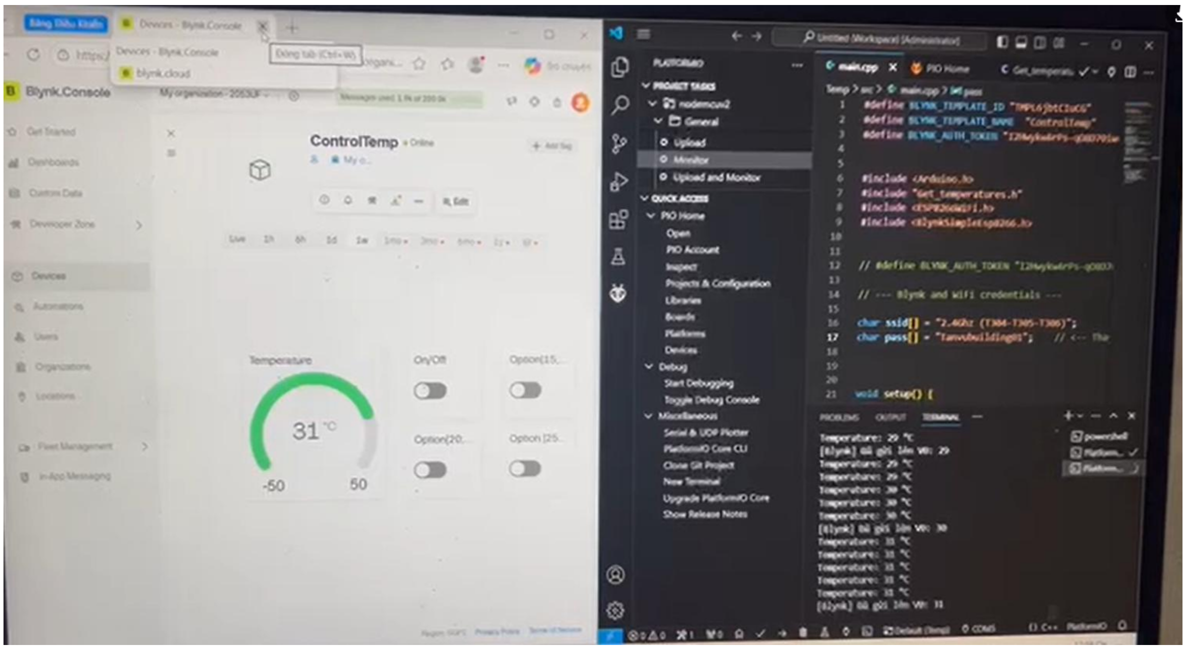



*Figure 18: Blynk dashboard template*

#### 7.2.3.3. Task 3.3 - Bi-Directional Synchronization Logic

A major challenge in the case of IoT technology is synchronizing the physical device and the app. This function requires the implementation of state logic. The ESP8266 keeps track of the mode as it receives it from the Arduino in the polling cycle. When the physical button on the fridge is pressed to change the mode, the ESP8266 receives the change in the mode and changes the virtual switches of the Blynk app accordingly using Blynk.virtualWrite().

### 7.2.4 Objective 4: System Integration & Testing

*Table 5: The list of testing components.*

| No | Testing component | Description | Date of successful testing |
|---|---|---|---|
| 1 | Temperature Sensor | Verify the accuracy of the DS18B20 sensor and the communication function with Arduino's 1-Wire interface. | 25/12/2025 |
| 2 | Cooling Unit | Connect the Peltier modules and Fans to the MOSFET driver for the purpose of confirming the cooling effect. | 25/12/2025 |
| 3 | User Interface (Local) | Try the LCD 16×2 to display real-time data and observe the presence of physical push buttons to respond to mode changes. | 25/12/2025 |
| 4 | IoT System (Wi-Fi) | Test the connectivity of the ESP8266 using the Blynk App for two-way data exchange with no latency for Read & Write operations. | 25/12/2025 |
| 5 | System Integration | Combine all subsystems into the refrigerator chassis. It then follows a stress test to ensure the specified temperatures (10−25 C) for 2 hours. | 31/12/2025 |

## 7.3. Proposed Model

On the basis of the system requirements and the final block diagram presented below, the proposed model is designed as a modular embedded system with the system placed in the SINO box that is fixed to the refrigerator chassis.

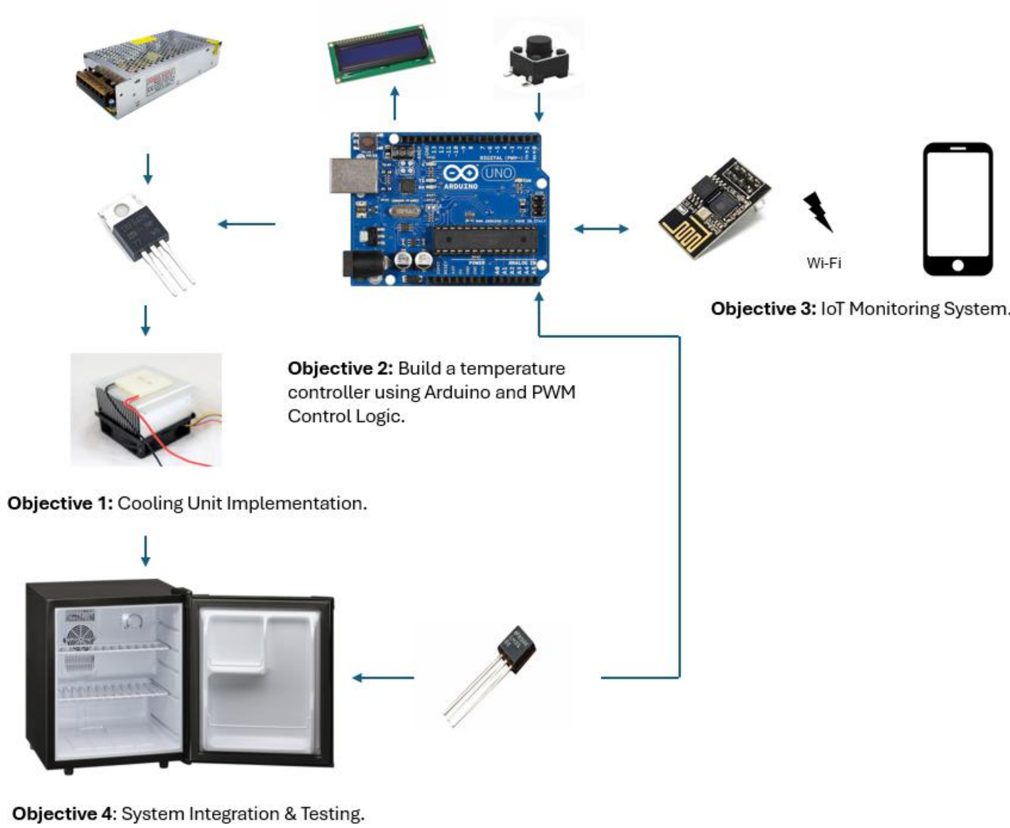



*Figure 19: System Block Diagram.*

The system has three primary subsystems that interact in the following manner:

1. **The Cooling Subsystem**: This component is situated on the side of the refrigerator and works by using the Peltier elements that are cooled by large DC fans. The subsystem operates with the help of the IRF540N MOSFET, which acts as an electronic switch.

2. **The Control Subsystem (or Slave)**: The Arduino Uno R3 is the local processing unit. It carries out the PWM algorithm to vary cooling power according to feedback from the DS18B20 temperature sensor inside the cold chamber.

3. **IoT Subsystem (Master)**: It uses ESP8266, which is the communication hub. It manages Wi-Fi communication with the Blynk server. The communication done by the

Master and Slave controls is done using the I2C protocol. The protocol provides a distinct separation of concerns regarding control operations versus network operations.

### 7.4. Flowchart

The system firmware is made to work in a non-blocking mode to respond to both physical and mobile app inputs. This system functionality flow can be indicated in the following Figure below.

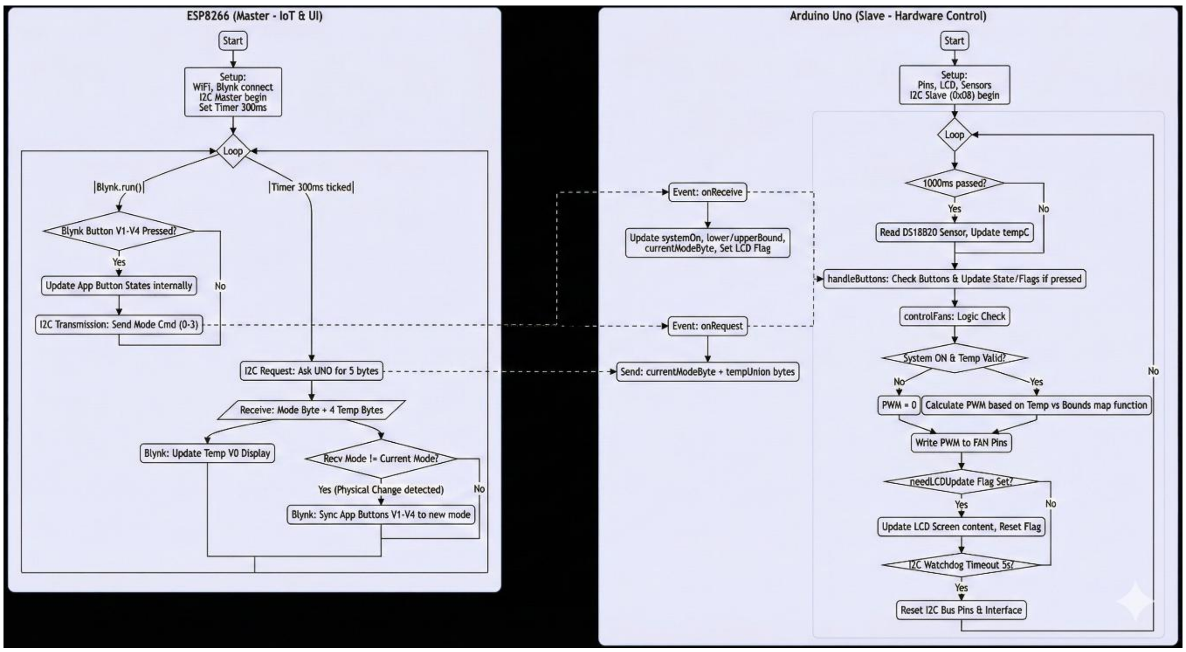



*Figure 20: System Operation Flowchart (ESP8266 Master & Arduino Slave).*

The Operational Logic Description unpacks

1. **Initialization**: Initialization of the LCD and the sensors by the Arduino along with the connection of the Wi-Fi network and the Blynk cloud by the ESP8266.

2. **Control Loop (Arduino)**: The Arduino continuously monitors the temperature. When the temperature is out of the range chosen by the user _, the Arduino uses the PID/PWM algorithm to determine the correct power output level (0 to 255) to be sent to the MOSFET.

3. **Communication Loop (ESP8266)**: The ESP8266 connects to the Arduino via I2C communication every 300ms to obtain the status information update. In the event of a change

of operation by the user via the App Blynk (V1-V4), the signal is transmitted instantaneously to the Arduino for the physical process adjustment.

## 7.5. Data Structure

To enable efficient data transfer from the ESP8266 (Master) to the Arduino Uno (Slave) via the I2C protocol, a specific byte-level data transfer protocol was developed between the devices. Rather than transferring data as plain information, the system uses packets for two-way data transfer:

### 7.5.1 Control Command Packet (Master → Slave)

It gets sent when the ESP8266 sends a write command for a change in the operational state of the Arduino. It carries a single byte: a uint8_t representation for the following logic:

- **Value 0**: Disable the system (OFF Mode).
- **Value 1**: Active cooling range = 10-15 C.
- **Value 2**: Active cooling range should be fixed to 15-20 degrees.
- **Value 3**: Set active cooling range to 20-25 C.

### 7.5.2 Status Feedback Packet (Slave -> Master)

Upon receiving the data request message sent by the Master, the Arduino transmits back a constant 5-byte packet comprising real-time data. The format is as follows:

Byte 0 (Operational Mode): Current operational mode or physical state of the controller. Used to match the interface on the mobile app to the buttons on the controller.

Bytes 1-4 (Temperature Data): This is the data representing the temperature, which is collected through the DS18B20 sensor. As the temperature is a floating-point value (Standard IEEE 754), it is broken down into individual bytes using the Union structure in the C++ code when it is sent. This allows the temperature data to be reconstructed back to its original value on receiving it through the ESP8266.

# CHAPTER VI – RESULTS

In this part, we demonstrate the effect or outcome or realization of our project, "Smart Temperature Controlled Mini-Fridge," by implementing the same in practical working conditions.

## 8.1. Results

### 8.1.1 Hardware Prototype and Circuit Design

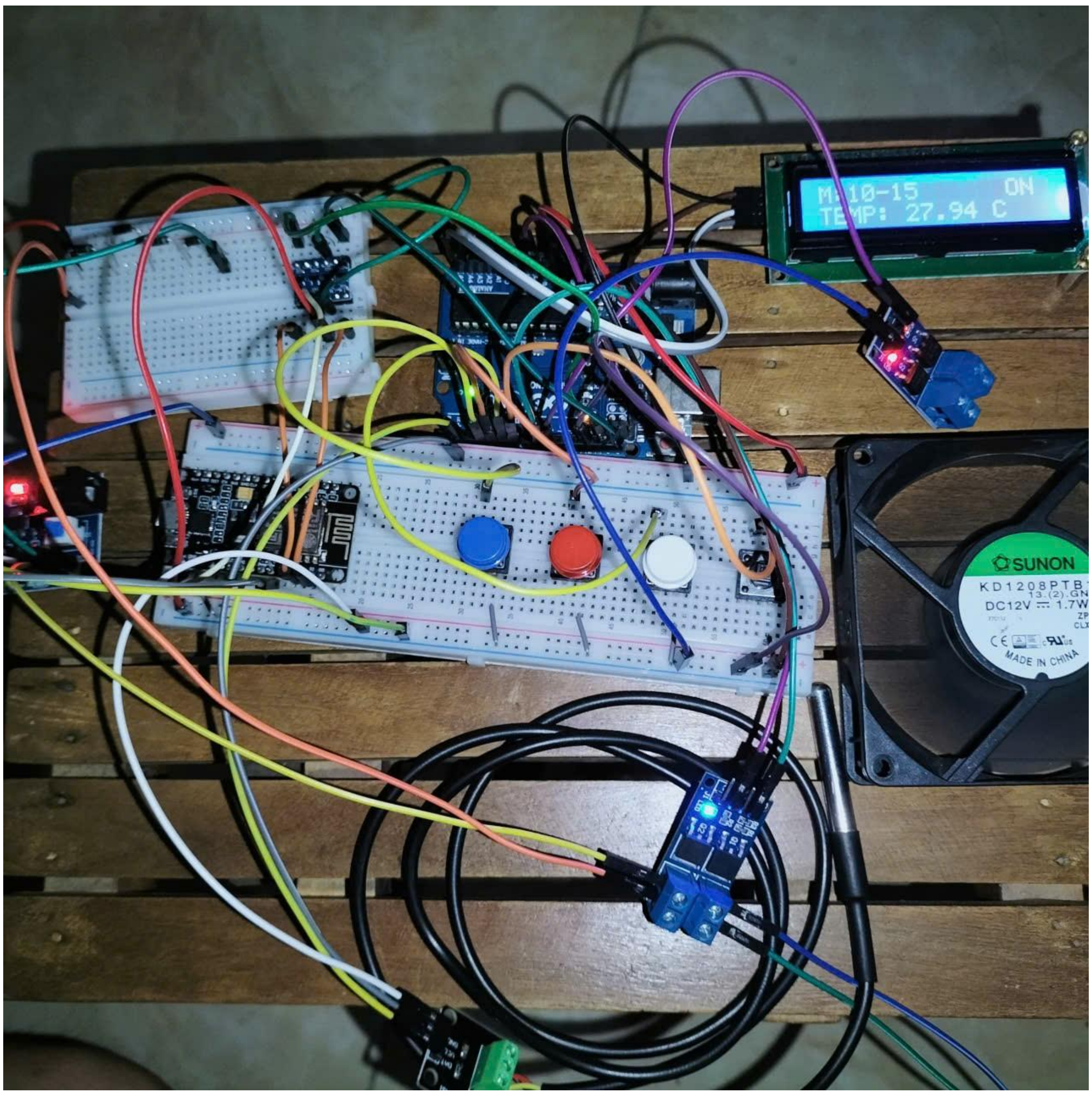



*Figure 21: The experimental setup of the control circuit on a breadboard.*

This image shows the testing phase on the breadboard. The system involves the integration of the Arduino Uno (Slave) and ESP8266 (Master) with the DS18B20 temperature sensor, LCD 16x2, and cooling fan. The LCD shows the real-time temperatures (for example, 27.94°C) as well as the current operating condition ("ON"). The check of the wiring at the stage helped to make sure that the logic signal for the MOSFET driver as well as the I2C communication protocol worked properly.

### 8.1.2 Integrated System (Final Product)


*Figure 22: The completed project integrated into the chassis (Side and Front view).*

The above image is a representation of the fully integrated mini-fridge. The casing of the mini-fridge is protected with yellow thermal insulation foam to curb heat emission. The system uses a professional electrical enclosure box referred to as SINO box to accommodate the central processing units of power for proper electrical safety and aesthetics. The cooling system component, such as the heatsink and fans, is fixed on the side wall of the mini-fridge to promote proper airflow.

### 8.1.3 IoT Interface (Mobile and Web App)

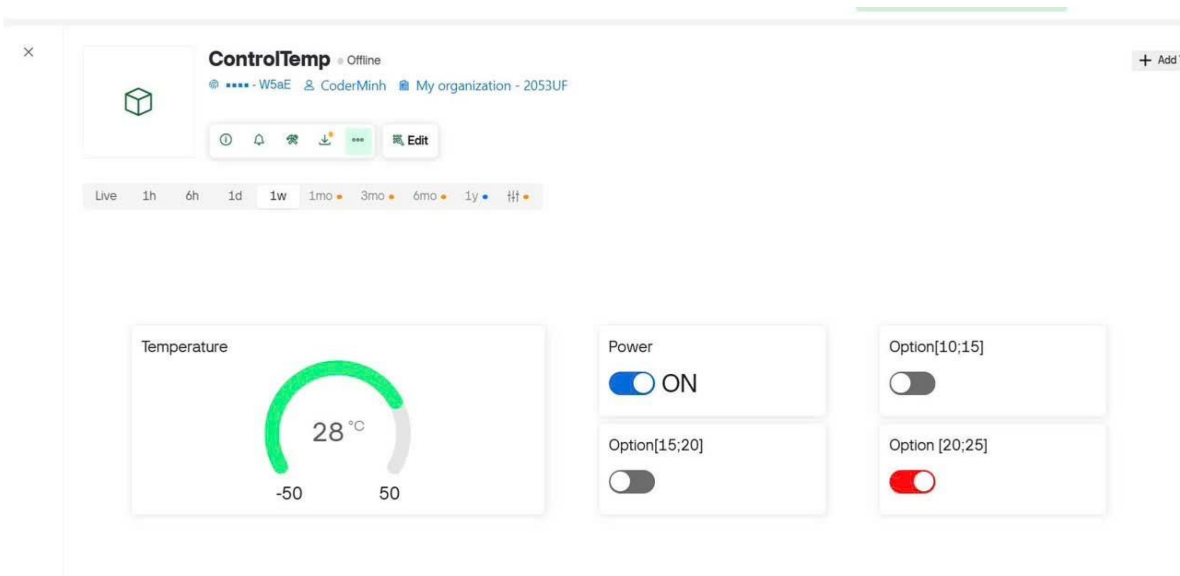



*Figure 23: The Blynk Web Dashboard for desktop monitoring.*

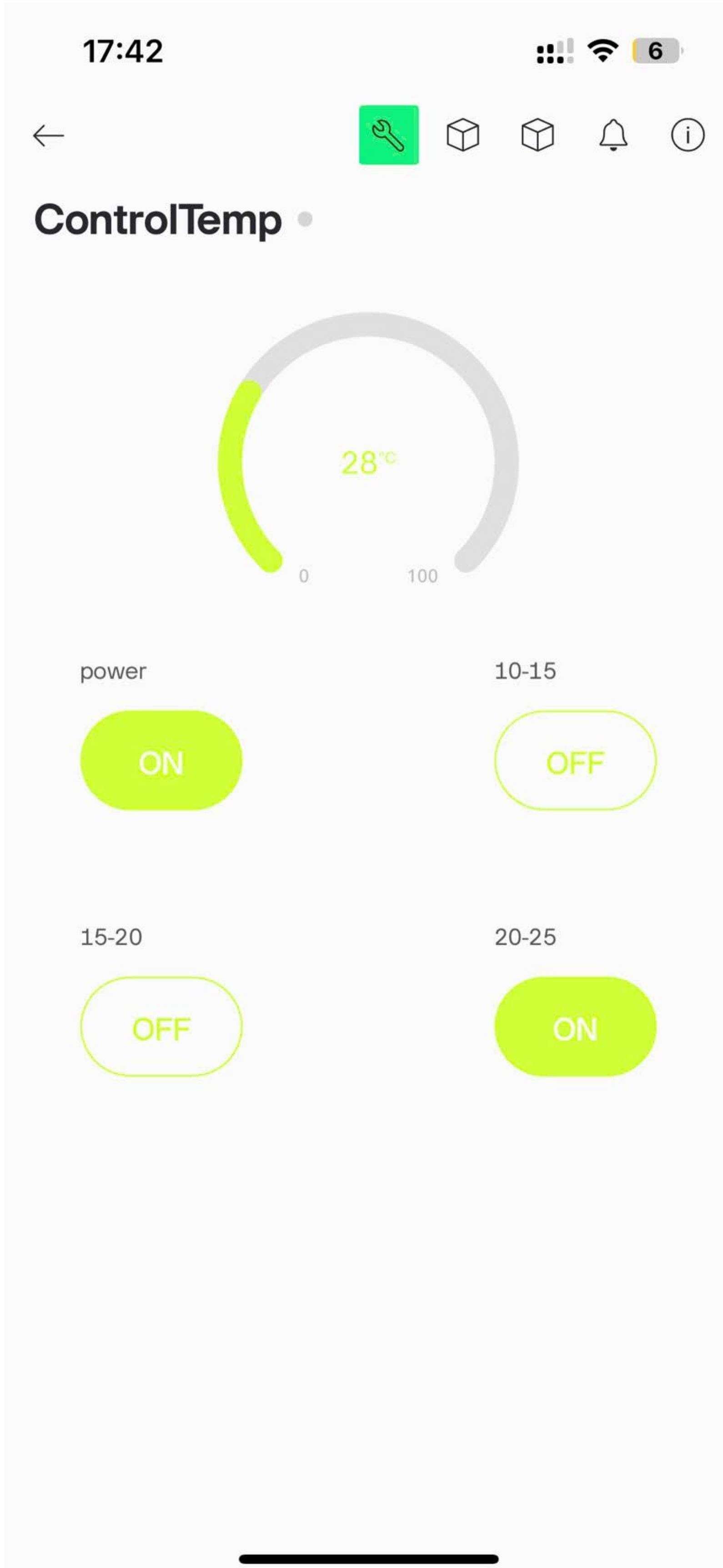

17:42
6
ControlTemp
28°C
0
100
power
ON
10-15
OFF
15-20
OFF
20-25
ON

*Figure 24: The Blynk Mobile App interface controlling the system.*

The remote monitoring feature of the system is tested using the Blynk platform. From first picture above, the mobile app consists of a gauge widget showing the actual temperature (28°C) and buttons representing the various modes of operation (Power ON/OFF, Range 10-15, 15-20, 20-25). From the next picture, it can be seen that the Web Dashboard is perfectly synchronized with the mobile app such that one can control the fridge using a variety of platforms.

### 8.1.4 Real testing result

*Table 6: Cooling Performance Test Data (Mode 10-15°C).*

| Time (Minutes) | Temperature (C) | Fan Speed (PWM) | Status |
|---|---|---|---|
| 0 | 28.0 | 0 | System Start |
| 10 | 24.5 | 255 (Max) | Cooling Down |
| 20 | 21.0 | 255 (Max) | Cooling Down |
| 30 | 18.0 | 255 (Max) | Cooling Down |
| 45 | 15.0 | 180 | Target Reached |
| 60 | 14.5 | 100 | Maintaining |

The system was tested in ambient temperatures of 28°C. The target mode was set to 10-15°C. The results show that the cooling module took around 45 minutes to cool the internal temperatures to the target set points. After the temperatures dropped to 15°C, the system automatically decreased the speed of the fans as per the PWM control logic, proving the validity of the control algorithm.

## 8.2. Discussion

### 8.2.1 Selection of Cooling Mechanism

After comparing different cooling methods, the Peltier effect-thin Thermoelectric [2] Cooler-was chosen because of its compact size, no moving parts (vibration-free), and being easily controllable with electronic signals. While it is not as powerful as a compressor, results will show that it should be sufficient in a mini-fridge application by maintaining temperatures between 10 C and 25 C effectively.

### 8.2.2 Control Logic and Power Efficiency

The use of PWM control as a means of Pulse Width Modulation as a replacement for the ON/OFF thermostat has proven effective. As has been observed during the testing stage, the circuit does not shut down the power fully once the temperature has been attained but reduces the duty cycles. This avoids the possibility of the Peltier element undergoing a shock due to the extreme variation in temperature.

### 8.2.3 IoT Synchronization and Latency

Master-Slave system implemented with the help of I2C communication protocol between ESP8266 and Arduino ran perfectly. The time difference from the pressing of the physical button on the fridge to the status update on the Blynk App took less than 1 second. This verifies that the division of the necessary control tasks (Arduino) and the communication tasks (ESP8266) is implemented perfectly to avoid system locking.

### 8.2.4 Power Consumption and Battery Feasibility Analysis

Although the currently used prototype operates on a grid-connected 12V-30A Industrial Switching Power Supply for the sake of stability in performance regarding the dual-core cooling system, an analysis of power consumption and possible battery operation follows next with the purpose of addressing the requirements for portability.

The system utilizes two high-power thermoelectric modules to achieve rapid cooling. The maximum power consumption (P_max) at peak load (100% duty cycle) is calculated as follows:

$$P_{\{max\}} = P_{\{Peltier1\}} + P_{\{Peltier2\}} + P_{\{Fans\}} + P_{\{Control\}}$$

- **Peltier Module 1 (12V-10A):** $12V \times 10A = 120W$
- **Peltier Module 2 (12V-12A):** $12V \times 12A = 144W$
- **Dissipation Fans & Controller:** 24W (Estimated for high-speed cooling fans)
- **Total Peak Power:** $\approx$ **288**W (Current draw $\approx 24A$)

### 8.2.5 Iterative Decision-Making Process

*Table 7: Table of discussion result.*

| Objectives | Problems | Iterative Process | Decision Making | Achieved/ Overcome | Improvements needed |
|---|---|---|---|---|---|
| **Objective 1:** Cooling Unit | **Problem 1:** Inefficient heat dissipation (Peltier hot side getting too hot, cold side not cooling). | **Approach 1:** Using a passive aluminum heatsink only. **Approach 2:** Adding a high-RPM fan and applying high-quality thermal grease. | **Approach 2 is selected.** Passive cooling was insufficient for the Peltier TEC1-12706, leading to thermal saturation. Active cooling is mandatory. | **Success:** The hot side temperature stabilized, allowing the cold side to drop to 10°C. | Better insulation for the fridge box to hold temperature longer. |
| **Objective 2:** Controller & Logic | **Problem 2:** Temperature fluctuation (Overshoot/ Undershoot). | **Approach 1:** Bang-Bang Control (Simple ON/OFF Relay). **Approach 2:** PWM Control using MOSFET and Mapping algorithm. | **Approach 2 is selected.** ON/OFF control caused the temperature to swing by ±3°C. PWM allows smooth power adjustment (slowing down the fan as it gets | **Success:** Temperature stability improved to within ±0.5°C of the setpoint. | Implement full PID (Proportional-Integral-Derivative) for even higher precision. |

| | | | | | |
|---|---|---|---|---|---|
| | | | closer to target). | | |
| Objective 3 | **Problem 3:** Data synchronization lag between App and Device. | **Approach 1:** Serial Communication (UART) between ESP & Arduino. **Approach 2:** I2C Protocol (Master-Slave) with Struct Data Packet. | **Approach 2 is selected.** UART conflicted with the USB debugging port. I2C proved more robust for multi-byte transfer (Float values). | **Achieved:** Real-time updates on Blynk with less than 1s latency. | Add a "Re-connect" feature if Wi-Fi signal is lost unexpectedly. |

# CHAPTER VII - CONCLUSION AND FUTURE WORK

## 9.1. Conclusions

In this project, it is shown that it is possible to successfully implement and manufacture an intelligent mini-fridge based on the Peltier effect with an IoT monitoring system. The first aim of designing and implementing an innovative solid-state cooling system with higher efficiency by replacing the traditional compressor cooling method using PWM logic has been fulfilled. From experiments conducted on the system designed, it has been shown that it is capable of maintaining prespecified temperatures (10-25° C) with higher stability. Moreover, it can also verify that there is perfect synchronization between the intelligent mini-fridge and its mobile app controlled by Blynk. Currently, it seems that there is some limitation to cooling capabilities due to the thermal inertia effect during use.

## 9.2. Future work

For further improvement of the functionalities and uses of this product, a number of suggested developments are proposed below:

1. **Algorithm Optimization**: Enhance the existing control algorithm by incorporating a complete PID control scheme that would eliminate error at the final stage.
2. **Hardware Enhancement**: A two-stage Peltier cooling system or an improved insulation material for the case needs to be implemented to lower the temperature below 5 ∘C with better energy efficiency.
3. **Portability**: Incorporate a rechargeable battery pack or input to enable the fridge to be fully portable, especially for camping or transporting patients.
4. **Smart Features**: Implement a feature to estimate cooling down time based on the prevailing ambient temperatures and send notifications to the client once the target temperature has been achieved.

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